

30538 Problem Set 4: Web Scraping

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Due 02/07/2026 at 5:00PM Central

“This submission is my work alone and complies with the 30538 integrity policy.” Add your initials to indicate your agreement: **AD**

Github Classroom Assignment Setup and Submission Instructions

(leave unchanged)

(40%) Step 1: Develop initial scraper and crawler

Scraping: Go to the first page of the HHS OIG's "Enforcement Actions" page and scrape: - Title - Date - Category - Link

Collect into a tidy dataframe and print `head`.

Answer

I scraped the first page of the HHS OIG Enforcement Actions webpage using Python with the `requests`, `BeautifulSoup`, and `pandas` libraries. The scraper extracted the title, date, category, and link for each enforcement action listed on the page.

Each enforcement action entry was parsed from the webpage and stored in a tidy dataframe with columns for Title, Date, Category, and Link. I verified the scraper output by printing the first few rows of the dataframe using `df.head()`.

(40%) Step 2: Making the scraper dynamic

a. Pseudo-code

Answer

The scraper function takes month and year as inputs.

First, the function checks whether the year is less than 2013. If so, it prints a message and stops execution.

Next: - Initialize an empty list for enforcement actions - Set page counter to 1

A `while` loop is used because the total number of pages is unknown.

For each page: - Construct URL using page number - Request webpage - Parse HTML - Extract title, date, category, and link - Append results - Increment page counter - Wait one second using `time.sleep(1)`

After scraping: - Convert results to dataframe - Save CSV file - Return dataframe

b. January 2024 results**Answer**

Running the scraper starting from January 2024 collected enforcement actions across multiple pages. The earliest enforcement action scraped corresponds to an entry from January 2024 listed under Criminal and Civil Actions.

c. January 2022 results**Answer**

Running the scraper starting from January 2022 collected enforcement actions across many pages. The earliest enforcement action in the dataset corresponds to an entry from January 2022. The dataframe from this run was saved to a CSV file and used for plotting.

(20%) Step 3: Plot data based on scraped data**Plot enforcement actions over time****Answer**

Using the dataset created in Step 2, I aggregated enforcement actions by month and year and created a line chart using Altair showing the number of enforcement actions since January 2022.

Plot categorized enforcement actions

Answer

First, I created a visualization comparing: - Criminal and Civil Actions - State Enforcement Agencies

Next, I manually classified enforcement actions within the Criminal and Civil Actions category into five topics:

- Health Care Fraud
- Financial Fraud
- Drug Enforcement
- Bribery/Corruption
- Other

Classification examples: - “health”, “medicare”, “fraud” → Health Care Fraud - “bank”, “financial” → Financial Fraud - “drug”, “opioid” → Drug Enforcement - “bribery”, “corruption” → Bribery/Corruption - remaining → Other

Altair was used to visualize topic trends over time.